

YUGESHKARAN P

Full Stack AI Engineer · GenAI & Agentic Systems · LangChain · CrewAI · RAG · PyTorch
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PROFESSIONAL SUMMARY

Full Stack AI Engineer with hands on experience in building and deploying production-grade agentic AI systems using Python, LangChain, CrewAI, and PyTorch. Hands-on expertise architecting multi-agent LLM workflows, RAG pipelines, and secure API backends with measurable production outcomes. Experienced in bridging AI prototypes to enterprise-grade deployments — implementing guardrails, rate limiting, JWT/OAuth authentication, Redis caching, and CORS security at the system level. Adept at co-building with technical teams and converting real-world AI integration challenges into scalable reusable solutions.

TECHNICAL SKILLS

Agentic AI & LLMs: LangChain, LangGraph, LangSmith, CrewAI, Savram, ReAct, multi-agent orchestration, MCP, tool-calling, Gemini

ML & Deep Learning: PyTorch, HuggingFace Transformers, NLP, RAG pipelines, embedding models, LLM inference optimization

Languages: Python, JavaScript (ES6+), TypeScript, SQL, C — data structures, algorithms, software design

Backend & Security: Flask, Node.js, Express.js — REST APIs, JWT/OAuth, rate limiting, CORS, Redis caching, guardrails

Cloud & DevOps: AWS (S3, Lambda, Amplify), Docker, Git/GitHub, CI/CD, Linux/Unix, Vercel

Databases & Vectors: MySQL, MongoDB, Redis, Pinecone VectorDB — semantic retrieval, schema design, query optimization

WORK EXPERIENCE

Bytes Base — Founder [BytesBase.me](#) [LinkedIn](#)

Sep-2025 - Present

AI Assisted Developer Community Platform for Academic Institutions. Designed for students, alumni and faculties to collaborate and grow together. It aims to foster Tech culture inside the academic institutes specially in tier -3 colleges.

- Architected and developed Bytes Base, using microservices architecture with Node.js and Flask services, enabling modular deployment and independent service scaling.
- Implemented **Redis server-side caching** on content feed API achieving **80% reduction in API calls** per user session, replaced session-based auth with **JWT authentication** across all client requests, eliminating session-store performance bottlenecks.
- Engineered API-level **rate limiting middleware** preventing **100% of abusive requests and DoS attack vectors**, built real-time Socket.io discussions, AWS S3 media storage, and deployed on Vercel with CI/CD practices.
- Built responsive React.js/TailwindCSS frontend and Framer motion, deployed on Vercel with CI/CD practices and maintained high availability under concurrent user load.

INTERNSHIP EXPERIENCE

Full-Stack Web Development Intern · Quantum Sharq Innovative Solutions

2024

- Built and optimized React.js/TailwindCSS web applications for clients, developed modular HRMS components in an Agile team, maintaining code quality through reviews and testing.

Data Science Intern · Comp Allah IT Industry

2023

- Performed data analysis, ML model training, and Python diagnostic scripting on business datasets, built visualizations to drive data-informed decision-making workflows.

RESEARCH PUBLICATIONS

- (Jun 2025)** "Evaluating the Performance of LLMs on SQL and NoSQL Databases Using LangSmith" — Intl. Conf. Multidisciplinary Solutions for Emerging Trends 2025
- (Sep 2024)** "Design and Development of an Attendance Management and Analysis System Using LLM" — ICIAAIEA 2024

PROJECTS

Assistant Knowledge Hub — Production RAG System [Github](#)

Python · Flask · OpenAI · Pinecone VectorDB · LangChain · Pydantic

- Built a production-grade RAG assistant with OpenAI text-embedding-3-small and Pinecone VectorDB, enforced **20 requests/minute/user rate limiting** to optimize LLM API costs and ensure equitable resource allocation across concurrent users.
- Implemented **CORS middleware blocking 100% of cross-site request forgery** attempts and deployed **LLM guardrails preventing 100% of prompt injection attacks** — delivering enterprise-grade security for a production agentic deployment.

Indic Research Scholar — Production RAG System [Github](#)

Flask · Sarvam Models · Pinecone VectorDB · Node.js · MongoDB · React.js · TailwindCSS

- Full-stack RAG platform helps researchers to query academic papers in 8+ Indian languages, with OCR ingestion and voice I/O via Sarvam AI (105B LLM, Saaras v3 STT, Bulbul v3 TTS).
- Built end-to-end auth (JWT + GitHub OAuth), multi-tenant document isolation, and a synchronous.

Mini Language Model — Custom NLP Training [Github](#)

Python · PyTorch · Transformers · Flask

- Designed and trained a 2.96M-parameter transformer-based LLM from scratch using PyTorch.
- Built an end-to-end pipeline covering tokenization, architecture design, optimization, and inference speed and memory efficiency tuning and exposed via Flask inference API.

EDUCATION

B.Tech — Artificial Intelligence & Data Science · Dhanalakshmi Srinivasan University, Trichy

2021 – 2025

CGPA: **9.21 / 10.0** · Class XII (CBSE): 92%